

SHANKAR BALACHANDRAN

ERP & Cloud Platform Architect · Openbravo / Etendo · AWS · Multi-Tenant SaaS
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PROFILE

Platform architect with 16 years in enterprise systems, effectively all of it in the Openbravo/Etendo ecosystem — spanning core product engineering, implementation, customer success, and platform migration. Two of those years were inside Openbravo's own core engineering team, on a product deployed across 1,000+ enterprise installations. Principal architect and sole engineer on a commercial cloud ERP (SaaS) taken live with two enterprise pilot clients and architected for multi-site rollout — designing, building and operating every layer of it single-handed, from the AWS production estate and enterprise SSO through the ERP back office to the mobile application framework and analytics stack. Earlier established and ran the Mauritius business unit of a French consultancy, building the delivery organisation to six engineers plus contractors to 15, with accountability for operations and P&L.

CORE COMPETENCIES

Cloud Architecture (AWS): EC2, Aurora PostgreSQL Serverless v2, S3, Nginx / SSL termination, Docker, CI/CD; elasticity, cost control, mount reliability, auto-pause tuning

Platform & Product Architecture: modular and loosely-coupled design, backward compatibility, reusable frameworks over one-off builds, long-lifecycle product foundations

Tenancy & Deployment: multi-tenant SaaS and single-tenant isolation from a single codebase with no per-client forks; isolated dev / UAT / production environments

Auth & Security: Keycloak SSO with OAuth2 / OIDC, silent login and cross-tab logout; Cloudflare Zero Trust (ZTNA) with OTP; rate limiting, CORS allowlisting, input validation, upload safety; passed enterprise security audit

API & Integration: versioned REST / SOAP contracts, web-service decorators, composite transactional endpoints, ERP-to-mobile orchestration, D365 invoice integration

Data & Analytics: PostgreSQL, PL/pgSQL, Cube.js pre-aggregated analytics decoupled from the transactional core, React dashboards

ERP Platform: Openbravo / Etendo core platform, Java, SmartClient, FlexDoc, modulescript frameworks, AD-level field and window customisation

Delivery: PMP and PMI-PMOCP certified; PMO governance, Agile/Scrum, SLA definition, vendor management

PROFESSIONAL EXPERIENCE

Founder & Principal Architect | Beaver minds Feb 2021 – Present

ERP modernisation and AI consultancy. Subcontracted to Openbravo's Customer Success unit for enterprise retail accounts (2021–2023), then principal engineering partner to AgilityERP (2024–present) across Openbravo-to-Etendo migrations, custom development and their customer portal — and from 2025, principal architect and sole engineer on TankSoft Pro, a commercial cloud ERP (SaaS) taken live with two enterprise pilot clients and architected for multi-site rollout. Worked directly with client executive teams throughout.

Cloud & Infrastructure Architecture

- **Designed and built the complete production AWS estate from scratch** — EC2, Aurora PostgreSQL Serverless v2, S3-backed attachments, Nginx with SSL termination, and GitHub CI/CD — across isolated DEV and UAT/production environments, enabling the enterprise pilot launch.
- **Owned cloud cost and reliability:** tuned Aurora auto-pause and background job scheduling to cut idle spend, and built an S3fs watchdog to guarantee attachment mount reliability.
- **Diagnosed and resolved Aurora-specific connection-pool exhaustion** — behaviour distinct from standard PostgreSQL — that was silently failing all background email processes; resolved via connection-pool minIdle tuning.
- Built and ran a three-tier estate covering the ERP back office, a Cube.js analytics service under Docker Compose, and a React dashboard front end.

Security Architecture

- **Implemented Cloudflare Zero Trust (ZTNA) end to end** — nameserver migration, tunnel deployment, access policies with email-OTP authentication across nine whitelisted domains, plus documented revert procedures. **Passed the enterprise security audit.**
- **Architected Keycloak single sign-on across two applications** — SPI jar, realm configuration, custom themes, token handoff, silent login, cross-tab logout, service-account authentication, and self-serve password reset.
- Delivered security-audit remediation across the platform: rate limiting, CORS allowlisting, input validation, and upload safety.

Platform & Product Architecture

- **Led Openbravo-to-Etendo platform migrations** across multiple client estates — including DCI, Indian River Transportation and Norac — covering assessment, module and customisation porting, and cutover, moving enterprise deployments onto a supported successor platform without rewriting them. Underpinned by two years inside Openbravo's own core engineering team.
- **Designed a reusable mobile-first application framework over Etendo**, shipping a Mobile Work Order wizard, a generic Manage Pictures attachment app, and a Sales Invoice Dispute app on shared foundations — a product line, not a set of one-offs. Field and operations staff completed real work from a phone under enterprise SSO during pilot.
- **Retired a legacy Spring Boot dispute portal**, replacing it with an in-platform dispute app covering the full status workflow, message threads, attachments and automated credit-note issuance — removing a manual back-office step entirely.
- Architected the end-to-end attachment stack — web-service decorator, AD configuration, S3 storage, regex filtering, and FlexDoc integration clickable from the back office — spanning desktop and mobile.
- Built a composite ManageOrder web service creating Sales Order and Work Order in a single call, plus Process Invoices and Create Credit Memo actions and dispute-to-credit automation.
- Built a real-time analytics module (React, Cube.js, PostgreSQL) with role-based access, deliberately decoupled from the transactional core so it evolves independently.

Scale & Outcomes

- **Owned every layer of the platform as sole engineer** — cloud infrastructure, security, ERP back office, mobile applications and analytics — across two production codebases, delivering six major subsystems in eight months through a disciplined dev → UAT → production pipeline.
- In pilot deployment, the unified Sales Order → Invoice → Work Order flow cut order cycle time from 30 minutes to 5 and removed 40% of duplicate data entry.
- Self-service portal built for 100+ external vendors and 25+ internal users, reducing manual dispute handling by 25%.
- Built the \$278K/year projected ROI business case that secured executive approval for the full multi-site rollout.
- Held the pilot stable while shipping: resolved mobile crash and hang loops, end-to-end timezone skew from device to database to PDF, cross-organisation deeplink loops, and a UAT data-source misconfiguration.

Earlier & Concurrent Engagements

- **Openbravo (now Orisha) — Customer Success unit, subcontractor, 2021–2023** — delivered customer success and product development for enterprise retail accounts including Decathlon, BUT (France) and Al Rubaiyat Group (MENA): enhancement roadmaps, escalation management, release planning and user enablement.
- **BUT Retail (France), Customer Success & Delivery Lead, 2024–2025** — reduced an active bug backlog from 100+ to under 20 through structured triage, and delivered a major platform version upgrade in under 60 days, restoring continuous delivery.
- **MCB Commercial Bank, Solution Architect, 2021** — designed an end-to-end Oracle data consolidation pipeline for RB5 transaction flow, achieving 85% consolidation and resolving terminal overpopulation.
- **Sharaf DG** — Openbravo/Etendo modernisation and integration work.

Technical Director & Acting Local Director | Kailao Consulting Limited — Ebene, Mauritius Oct 2014 – Feb 2021

Recruited by the founder of Kailao (France) to establish and run its Mauritius business unit; acting local Director accountable for operations and P&L.

- **Set up the Mauritius operation from scratch** and built the delivery organisation to six permanent engineers plus a contractor bench taking capacity to 15, across 5+ concurrent client programmes.
- Held P&L accountability as acting local Director, improving client contribution margins by 25% through resource allocation and programme prioritisation.
- Established PMO governance, delivery standards and reporting cadence from nothing; sustained 98% on-time, on-budget delivery.
- Led 6+ end-to-end Etendo/Openbravo implementations across retail, manufacturing and services — including supermarket POS, ERP and POS for a beverage business, and a manufacturing ERP rollout.
- Architected a real-time Xero-to-Openbravo accounting integration that removed 80% of manual reconciliation, and a Mauritius-compliant payroll/HRMS module reused across multiple clients.

Applications Engineer | Openbravo (now Orisha) — Bangalore, India Aug 2012 – Sep 2014

Core platform engineering team for a commercial ERP deployed across 1,000+ global enterprise installations.

- Designed and shipped core platform features and architectural improvements to scalability, stability and usability across a large Java/XML codebase.
- Reduced critical support tickets by 20% through proactive defect resolution.

- Published open-source contributions to the platform's preferences architecture and attachment model — foundations still carried by the product today.

Technical Consultant → Senior Technical Consultant | Fugo Consulting Pvt. Ltd. — Chennai, India *Dec 2009 – Aug 2012*

- Openbravo ERP module and vertical development, and HRMS product development — the start of a near-continuous five-year run on the Openbravo platform.
- Implementation lead for an IRDA-compliant insurance claims portal serving 200+ users and 300+ claims per day.
- Built a centralised TDS management system spanning 4,000+ hospitals with full regulatory compliance.

CERTIFICATIONS

- Project Management Professional (PMP) — Project Management Institute, 2020–2029
- PMO Certified Professional (PMI-PMOCP) — Project Management Institute, 2025–2028

EDUCATION

- Master of Computer Applications (MCA), Computer Science — Anna University, Chennai, 2010
- B.Sc. Mathematics — Loyola College, Chennai, 2007

ADDITIONAL

Languages: English, Tamil, Hindi (French — elementary, in progress). Work authorisation: India; Mauritius Self-Employed Occupation Permit valid to 2031. Lived and worked in Mauritius 2015–2024.